

Lecture 2 - Gauge theories on a lattice

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Abstract

In this lecture we begin by discussing the fate of spacetime symmetries on a discrete lattice. We then briefly remind ourselves of the continuum abelian gauge theory and gauge transformations. Following this we see that demanding gauge invariance for the scalar field theory (on the lattice) gives new structure in contrast to what is done in the continuum. We then discuss the basics of lattice gauge theories starting from defining links and plaquettes, building the Wilson gauge action. If time permits, we then end with explaining how to fix a gauge on the lattice using maximal trees, finally introducing some important observables like the Wilson and Polyakov loops. This lecture is based on many references including [1, 2, 3, 4].

1 Overview: Symmetries on the Lattice

As we have seen previously, the discrete spacetime lattice — with properly defined lattice fields and observables — provides us with a powerful tool to compute finite quantities whose correct continuum extrapolation leads to expected results. However, as one might expect, the discrete lattice explicitly breaks certain continuum symmetries, which must be recovered in the continuum limit.

We can broadly divide these symmetries into two classes: spacetime and internal (field). When considering spacetime symmetries, one should keep in mind that we are working in *Euclidean* space - obtained by Wick rotation from the Minkowski formulation - so that the relevant continuum symmetry group is $SO(4)$ rather than the Lorentz group $SO(3,1)$. This is a crucial simplification: Lorentz invariance, involving non-compact boosts, is *qualitatively* broken on any discrete lattice, a boosted time-slice generically misses all lattice sites, and there is no controlled sense in which the lattice approximates boost symmetry. By contrast, the full rotation group $SO(4)$ is broken only to the discrete hypercubic group $H(4)$, and the resulting symmetry-violating operators are irrelevant in the renormalization group sense - suppressed by powers of the lattice spacing a - so that $SO(4)$ invariance is systematically recovered in the continuum limit.

Internal symmetries, on the other hand, need not be affected by the discretization of spacetime at all. A simple example is the $\phi(x) \rightarrow -\phi(x)$ symmetry of the free scalar field, which holds identically on the lattice since it acts pointwise and does not involve derivatives or transport between sites. In this lecture, we will consider much more tricky example: what happens when we demand *local gauge invariance* on the lattice.

See David Tong's lecture notes[5] on Lattice Field Theory for a insightful treatment on broken Lorentz invariance on the (Minkowski) lattice.

This discussion on $H(4)$ symmetry is taken from Guy Moore's lecture[6] on LFT and is explained there in great detail.

2 Continuum Abelian Gauge Theory: A Recap

Before discussing how to make our theory locally gauge invariant, it is good to remind ourselves what gauge fields are and what action describes them - in the continuum. One should recall that the Standard Model of particle physics is made up of matter fields (like scalar and fermions) and force fields (photons and gluons). Quantum electrodynamics (QED) describes the interaction of fermions (electrons, muons so on) with photons which are described by an *abelian gauge field theory*. While quantum chromodynamics (QCD) describes the dynamics of quarks and gluons, where the gluons are described by *non-abelian gauge fields*.

Due to the mathematical simplicity of abelian gauge fields, we will restrict ourselves in this lecture to studying those. Euclidean abelian gauge field theory is described by the following *Yang-Mills* action,

$$S[A] = \frac{1}{4} \int d^4x F_{\mu\nu}(x) F_{\mu\nu}(x), \quad (1)$$

where $F_{\mu\nu}(x) = \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x)$ is the anti-symmetric field strength tensor and $A_\mu(x)$ is the vector potential. The field strength tensor $F_{\mu\nu}(x)$ has the property that it is invariant under the following *gauge* transformation,

$$A'_\mu(x) = \Omega(x) A_\mu(x) \Omega^\dagger(x) - i\Omega(x) \partial_\mu \Omega^\dagger(x), \quad (2)$$

when $\Omega(x)$ are elements in the abelian group $\mathcal{G} = U(1)$. These are then just phase factors and commute with each other, $\Omega(x)\Lambda(x) = \Lambda(x)\Omega(x)$ and have the property that $\Omega^\dagger(x) = \Omega^{-1}(x)$. Choosing such a phase factor to be $\Omega(x) = e^{i\alpha(x)}$, and using the abelian property (which allows us to move these phase factors past each other), Eq. (2) becomes,

$$A'_\mu(x) = A_\mu(x) + \partial_\mu \alpha(x). \quad (3)$$

The transformation represented by Eq. (3) is what is usually known as *local gauge transformation* and you might remember this from your course in Electrodynamics, as these transformations leave physical observables like electric and magnetic fields invariant. As can also be easily checked, this transformation also leaves the action $S[A]$ invariant.

This gauge invariance introduces a problem when we try to naively construct the path inetgral for the gauge theory. To see this, notice that for every configuration $A_\mu(x)$, there exist infinitely many gauge-equivalent configurations related by Eq. (3) that give identical physics - the same action, the same observables. These gauge-equivalent configurations form an *orbit of the gauge group* $\mathcal{G}$. The measure $\mathcal{D}A_\mu$ integrates over all configurations, including all copies within each orbit, so it factorizes as

$$\mathcal{D}A_\mu = \mathcal{D}A_\mu^{\text{phys}} \cdot \mathcal{D}\alpha, \quad \mathcal{D}\alpha = \prod_x d\alpha(x), \quad (4)$$

where $\mathcal{D}A_\mu^{\text{phys}}$ is the measure over gauge-inequivalent configurations and $\mathcal{D}\alpha$ is the measure over the gauge group. Since the action is gauge invariant, the integral over $\mathcal{D}\alpha$ yields only the volume of the gauge group,

$$Z = \text{Vol}(\mathcal{G}) \int \mathcal{D}A_\mu^{\text{phys}} e^{-S[A]}, \quad (5)$$

You may notice that the Euclidean Yang-Mills action comes with a positive sign as opposed to the Minkowski version.

which diverges since $\alpha(x) \in \mathbb{R}$ at every spacetime point, making $\text{Vol}(\mathcal{G}) = \infty$. This infinite overcounting must be dealt with - in the continuum this is achieved via the Faddeev-Popov procedure¹.

We will see below that on the lattice, gauge invariance is built in exactly and the problem is resolved differently: the link variables $U \in SU(N)$ become the fundamental degrees of freedom instead of the vector potential and take values in a compact group, so $\text{Vol}(\mathcal{G})$ is finite and the path integral is naturally regulated without the need for gauge fixing.

3 Local Gauge Invariance in the Lattice Free Scalar Field Theory

In this section we will begin by considering some internal symmetries of the scalar field theory given by a complex scalar field $\phi(x) = (\phi_1(x) + i\phi_2(x))/\sqrt{2}$, whose (Euclidean) action is given by,

$$S_E[\phi] = \int d^4x \left\{ \partial_\mu \phi^*(x) \partial_\mu \phi(x) + M^2 |\phi|^2(x) \right\}. \quad (6)$$

Looking at the action, it is easy to see that, a rotation by a global phase $\phi(x) \rightarrow e^{i\alpha} \phi(x)$, leaves the action invariant.

However, if one wants to couple the free scalar field with gauge fields, then to have a consistent theory, the entire action must respect local gauge invariance as discussed in the previous section. The way gauge invariance acts on matter fields is by applying local, space dependent phases to the fields: $\phi(x) \rightarrow e^{i\alpha(x)} \phi(x)$. One immediately sees that under this transformation, the kinetic term of the free scalar field is not invariant and transforms as,

$$\partial_\mu \phi'(x) = e^{i\alpha(x)} (\partial_\mu \phi(x) + \partial_\mu \alpha(x)). \quad (7)$$

The mass term on the other hand is unchanged under the transformation. The procedure to make the full action gauge invariant is by modifying the derivative term from $\partial_\mu \rightarrow D_\mu$, where $D_\mu = \partial_\mu + iA_\mu$. Assume that the gauge transformation is given by $A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu \lambda(x)$, then D_μ transforms as,

$$\begin{aligned} D_\mu \phi(x) &\rightarrow D'_\mu \phi'(x) \\ &= (\partial_\mu + iA_\mu + i\partial_\mu \lambda(x)) e^{i\alpha(x)} \phi(x) \\ &= e^{i\alpha(x)} (\partial_\mu + iA_\mu + i\partial_\mu \lambda(x) + i\partial_\mu \alpha(x)) \phi(x). \end{aligned} \quad (8)$$

In order to recover gauge invariance - we need to choose $\lambda(x) = -\alpha(x)$ giving us the following rules,

$$\begin{aligned} A_\mu &\longrightarrow A_\mu(x) + \partial_\mu \lambda(x), \\ \phi(x) &\longrightarrow e^{-i\lambda(x)} \phi(x), \\ D_\mu \phi(x) &\longrightarrow e^{-i\lambda(x)} D_\mu \phi(x). \end{aligned} \quad (9)$$

In the continuum, this would result in the following gauge invariant action,

$$S_E[\phi, A] = \int d^4x \left(D_\mu \phi^*(x) D_\mu \phi(x) + M^2 \phi^*(x) \phi(x) + \frac{1}{4} F_{\mu\nu}(x) F_{\mu\nu}(x) \right), \quad (10)$$

¹This is something you learn in your QFT or Advanced QFT courses

You can learn more about continuum scalar quantum electrodynamics in Chapter 9 of the book by M. Schwartz[4].

with the resulting path inetgral,

$$Z = \int \mathcal{D}A \mathcal{D}\phi e^{-S_E[\phi, A]}. \quad (11)$$

In order to construct a lattice version of the gauge invariant action for the interacting scalar and gauge field theory, we need to consider (i) what happens to the lattice derivative and more importantly, (ii) what observable should we choose to represent the gauge degrees of freedom on the lattice.

To begin the discussion, we should look at Eq. (9) again and realise that the second transformation should also hold on the lattice the phase factor just becomes a function of lattice sites giving,

$$\begin{aligned} \phi'(na) &= e^{-i\lambda(na)} \phi(na) \\ \phi'^*(na) &= \phi^*(na) e^{i\lambda(na)}. \end{aligned} \quad (12)$$

Considering the (dimensionless) lattice action from the previous lecture - now with a complex scalar field,

$$S_E = - \sum_{n, \hat{\mu}} \left(\hat{\phi}_n^* \hat{\phi}_{n+\hat{\mu}} + \hat{\phi}_{n+\hat{\mu}}^* \hat{\phi}_n \right) + (8 + \hat{M}^2) \sum_n \hat{\phi}_n^* \hat{\phi}_n, \quad (13)$$

it is easy to see that under a gauge trasnformation, only the *hopping* term of the action transforms non-trivially. We further use the notation of group elements $\Omega(n) = e^{-i\lambda(n)}$ to see,

$$\sum_{n, \hat{\mu}} \hat{\phi}_n^* \hat{\phi}_{n+\hat{\mu}} \rightarrow \sum_{n, \hat{\mu}} \hat{\phi}_n^* \Omega^\dagger(n) \Omega(n + \hat{\mu}) \hat{\phi}_{n+\hat{\mu}}. \quad (14)$$

It should be clear that $\Omega^\dagger(n) \Omega(n + \hat{\mu}) \neq 1$. This leads us to consider the parallel transporter or a Wilson line, which for an abelian gauge theory is simply,

$$U(x, y) = e^{ie \int_x^y \sum_\mu dz_\mu A_\mu(z)}. \quad (15)$$

The physical relevance of this quantity is that it tells us how to *transport* a field in gauge space from one spacetime point to another. Moreover, under a gauge trasnformation, it transforms like,

$$U(x, y) \rightarrow \Omega(x) U(x, y) \Omega^\dagger(y). \quad (16)$$

This gives us a hint on how to construct the lattice hopping term, such that gauge invariance is preserved. The quantity one should consider is $\phi^*(x) U(x, y) \phi(y)$, which remains invariant under a gauge transformation,

$$\begin{aligned} \phi^*(x) U(x, y) \phi(y) &\rightarrow \phi^*(x) \Omega(x) \Omega^\dagger(x) U(x, y) \Omega^\dagger(y) \Omega(y) \phi(y) \\ &= \phi^*(x) U(x, y) \phi(y) \end{aligned} \quad (17)$$

Hence, the gauge invariant action for the complex scalar field is given by,

$$S_E = - \sum_{n, \hat{\mu}} \hat{\phi}_n^* U(n, n + \hat{\mu}) \hat{\phi}_{n+\hat{\mu}} + \hat{\phi}_{n+\hat{\mu}}^* U(n + \hat{\mu}, n) \hat{\phi}_n + (8 + \hat{M}^2) \sum_n \hat{\phi}_n^* \hat{\phi}_n. \quad (18)$$

In the next section we will consider how these Wilson lines look on the lattice and how to construct the action for the abelian lattice gauge theory.

4 Gauge Fields on the Lattice: Links, Plaquettes and the Wilson Action

In lattice gauge theory, the fundamental gauge degrees of freedom are not the continuum vector potentials $A_\mu(x)$, but rather group-valued variables associated with the links and plaquettes of the lattice. These variables encode the parallel transport of fields and the curvature (field strength) in a gauge-invariant way.

Link Variables

On a hypercubic lattice, each site is labeled by an integer vector n and each direction by $\hat{\mu}$. The link variable $U_{n,n+\hat{\mu}}$ is an element of the gauge group and is associated with the oriented link from site n to site $n + \hat{\mu}$. For an abelian theory ($U(1)$),

$$U_{n,n+\hat{\mu}} = e^{iaeA_\mu(n)} \quad (19)$$

where a is the lattice spacing and e is the gauge coupling (keeping with the notation of Eq. (15)). As we have already seen, link variables act as parallel transporters for matter fields between neighboring sites. A pictorial representation of links can be seen in Fig. 1, where the link (blue) and its conjugate (red) are shown. Notice that since the links are directional, it makes sense to choose a convention for positive directions. For the rest of this lecture series we will choose up and right to be the links, with down and left representing the conjugates. This is why the links are colored in the figure.



Figure 1: Links: fundamental gauge degrees of freedom on the lattice

Before we move on to the next section, we will also adopt the following definition to represent a link (this is just following the conventions of the lattice community). In these conventions, $U_\mu(n) \equiv U_{n,n+\hat{\mu}}$ and $U_\mu^\dagger(n) = U_\mu^{-1}(n) = U_{-\mu}(n + \hat{\mu}) \equiv U_{n+\hat{\mu},n}$.

Plaquette Variables

The smallest nontrivial closed loop on the lattice is the plaquette, a square formed by four links in the μ - ν plane. The plaquette variable at site n is defined as

$$U_{\mu\nu}(n) = U_\mu(n)U_\nu(n + \hat{\mu})U_\mu^\dagger(n + \hat{\nu})U_\nu^\dagger(n) \quad (20)$$

For $U(1)$, this is just the product of four phases around the plaquette, shown in Fig. 2. The plaquette is a gauge invariant observable and can be shown to be related to the continuum field strength tensor as follows,

$$U_{\mu\nu}(n) \approx e^{ia^2 F_{\mu\nu}(n)}, \quad (21)$$

where the lattice field strength tensor is given in terms of differences of the vector potential,

$$F_{\mu\nu}(n) = \frac{1}{a} [A_\nu(n + \hat{\mu}) - A_\nu(n) - (A_\mu(n + \hat{\nu}) - A_\mu(n))] . \quad (22)$$

For non-abelian gauge theories like $SU(N)$, it is a product of matrices and one needs to take a trace over the product to recover gauge invariance. This is part of an exercise in the next sheet.

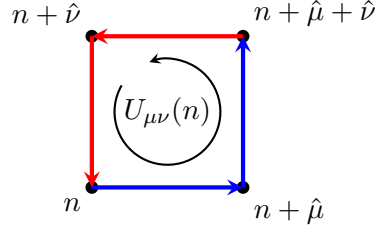


Figure 2: Plaquette: smallest gauge-invariant quantity constructed from links

The Wilson Action

The Wilson action is the standard gauge-invariant action for pure gauge theory on the lattice. To see how it can be constructed from plaquettes, one should look at Eq. (21) and recognize that an expansion of the exponential to third order gives us the term appearing in the continuum action in Eq. (1),

$$\frac{1}{e^2} \sum_n \sum_{\mu < \nu} \left[1 - \frac{1}{2} (U_{\mu\nu}(n) + U_{\mu\nu}^\dagger(n)) \right] \approx \frac{1}{4} \sum_{n,\mu,\nu} a^4 F_{\mu\nu}(n) F_{\mu\nu}(n), \quad (23)$$

where the term $1/e^2$, is lattice nomenclature is usually called the *inverse coupling* denoted by β . Also notice that that on the right-hand-side of Eq. (23), an extra factor of $\frac{1}{2}$ appears because the sum over directions now includes both positive and negative directions. Hence, the Wilson gauge action for abelian gauge theory is,

$$S_W[U] = \beta \sum_n \sum_{\mu < \nu} \left[1 - \frac{1}{2} (U_{\mu\nu}(n) + U_{\mu\nu}^\dagger(n)) \right]. \quad (24)$$

Typically in simulations, the constant volume term is dropped.

Although we did not derive the corresponding expressions for non-abelian gauge fields like $SU(N)$ in this lecture, it is worth noting that the form of the Wilson action, plaquettes terms remains similar with one important difference being that the links are not phases anymore but $N \times N$ matrices for $SU(N)$ and to ensure gauge invariance, just a product of links over a closed loop is not sufficient - we must take a trace over the plaquette. You are encouraged to go through these calculations using as references [1, 3, 2].

There is an exercise in the second sheet related to the lattice Wilson action and it's naive continuum limit.

The next exercise sheet also includes the treatment of non-abelian gauge theories.

5 The Lattice Path Integral for $U(1)$ Gauge Theory

Now that we have formulated the lattice abelian gauge theory completely in terms of compact variables like links and plaquettes, we can write down the path integral for the pure gauge theory,

$$Z = \int \mathcal{D}U e^{-S_W[U]}, \quad (25)$$

where the path integral measure is given by $\mathcal{D}U \equiv \prod_{n,\mu} dU_\mu(n)$.

In order to understand the properties of this measure, we begin by demanding that the path integral Z is a gauge invariant object. One can either argue this by remembering that physical quantities should be gauge invariant or by viewing gauge transformation as change of variables and recalling that integrals are invariant under change of variables. Since we know that the action is gauge invariant by construction, $S_W[U] = S_W[U']$, (with $U' \equiv U'_\mu(n) = \Omega(n)U_\mu(n)\Omega^\dagger(n + \hat{\mu})$), we require the path inetgral measure to be gauge invariant as well. This implies,

$$dU_\mu(n) = dU'_\mu(n) = d(\Omega(n) dU_\mu(n) \Omega^\dagger(n + \hat{\mu})). \quad (26)$$

An important aspect about gauge transformations is that one can *independently* choose a gauge trasnformation at each point in spacetime (or on the lattice). This gives a stronger condition on the invariance properties of the measure over the measure dU , i.e., the group element must be independently left and right multiplication invariant,

$$dU = d(VU) = d(UV). \quad (27)$$

One aspect that remains to be understood is: what is the volume associated with this measure? This requires a little discussion because this aspect is fundamentally different in the link formulation from what we encountered in Section 2. Restricting ourselves to formulating lattice gauge theories on links, restricts the integration domain of the vector potentials $aA_\mu \in \{-\pi, \pi\}$ (recall that $A_\mu(n)$ has dimension $\frac{1}{a}$). This gives us a normalization condition for the measure,

$$\int dU_\mu(n) = \int_{-\pi}^{\pi} \frac{d(aA_\mu(n))}{2\pi} = 1. \quad (28)$$

These conditions, with Eqs. (27) and (28), taken together, define the so-called *Haar measure*. This is a measure for integration over continuous compact groups. This completes our discussion on putting an abelian gauge theory on a lattice while preserving gauge invariance exactly.

Apprently, this was first introduced by A. Hurwitz in 1897 under the name "invariant integral" (source: Wikipedia)

6 Gauge Fixing via Maximal Trees

7 Important Observables: Wilson and Polyakov Loops

References

- [1] Heinz J. Rothe. *Lattice Gauge Theories: An Introduction*, volume 82 of *World Scientific Lecture Notes in Physics*. World Scientific, 4th edition, 2012.
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